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Two improved methods for low-speed hot-wire calibration

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Abstract. Two improved methods for hot-wire anemometer calibration in the low-speed range (15–95 cm/s) are described: (i) a laminar pipe-flow method, and (ii) a shedding-frequency method. For the laminar pipe-flow method the calibration is performed in the exit plane of a fully-developed laminar pipe flow. For the shedding-frequency method the shedding frequency of a parallel mode cylinder wake is measured; the velocity is then obtained from a continuous Strouhal–Reynolds number relationship. Calibration results from the present methods were compared with each other and with results from a commercially available calibration jet. The present methods were demonstrated as a simple, but accurate means for low-speed hot-wire calibration.

1. Introduction

Calibration of hot-wire sensors at low velocities poses problems because of the insensitivity of conventional calibration methods. Application of hot-wire anemometry at low velocities calls for special calibration techniques.

Different techniques for low-velocity hot-wire probe calibration have been developed. Commercial calibration jets are available, e.g., the TSI 1125 calibrator, which has a low-speed range from 2 to 90 cm/s. In this calibration jet the pressure drop from the plenum chamber to the jet exit is measured and used to obtain velocity from supplied graphs. In the low-speed range, the hot-wire must be placed in the plenum chamber of the calibration jet, where the velocity profile lacks the uniformity that is found at the jet exit. TSI quotes an accuracy of $\pm 5\%$ for velocities from 15 to 305 cm/s and $\pm 10\%$ for velocities below 15 cm/s. The calibration results of the TSI 1125 calibrator in the low speed range will be compared with the present results in section 4.

Kohan and Schwarz (1973) calibrated a hot-wire at low wind speed using the Roshko (1954) Strouhal–Reynolds number (SR) relationship for flow Reynolds number range between 50 and 150 ($S = f_s D / U$, where f_s is the shedding frequency, U the velocity and D the cylinder diameter). The velocity was obtained by measuring the vortex-shedding frequency and then using the SR relationship. However, the vortex-shedding mode behind the test cylinder was not parallel in the experiments of Kohan and Schwarz. Calibrations performed with non-parallel vortex shedding can lead to significant inaccuracies as will be described in section 3.

Aydin and Leutheusser (1980), and Tzanis (1987) produced laminar, plane Couette flow to calibrate a hot-wire at very low velocities. Christman and Podzimek (1981) calibrated a hot-wire using the nozzle of a DISA 55D41/42 calibrator. The flow in Christman and Podzimek's experiment was generated by draining water at a controlled rate from an air-tight tank connected to the DISA nozzle. Manca *et al* (1988) investigated the calibration results of hot-wire probes for low air velocities with variable temperature. The hot-wire was placed at the exit of a glass tube in which a Hagen–Poiseuille flow was established. Brunn *et al* (1989) used a swing-arm apparatus to calibrate hot-wire probes at low velocities. However, all of the preceding methods are limited in one or several of the following aspects: (i) the complexity and expense of the apparatus, (ii) the lack of confirming results by another independent calibration method, and (iii) the inaccuracy of the method itself.

In the present study, two improved low-speed hot-wire calibration methods were developed for wind speeds ranging from 15 to 95 cm/s: (i) the laminar pipe-flow method, and (ii) the shedding-frequency method. Calibration results from the two improved methods were compared with results from the TSI 1125 calibrator.

2. The laminar pipe-flow method

The apparatus used for the low-speed hot-wire calibration is a modified TSI 1125 calibrator. Figure 1 shows the apparatus for obtaining the low velocity measurements. The heat exchanger and the plenum chamber/contraction of a TSI 1125 calibrator were used as part of the calibration apparatus to ensure constant temperature and uniform flow at the entrance of the glass

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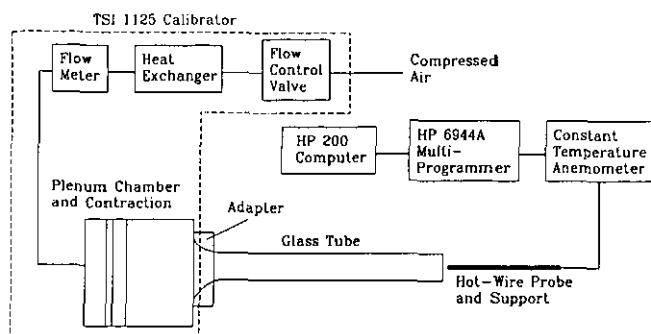


Figure 1. Pipe flow apparatus for the calibration of hot-wire anemometer at low wind speeds. The glass tube and the hot-wire probe were oriented vertically in the laboratory.

calibration tube. A glass tube was installed at the exit of the contraction of a TSI 1125 calibrator by means of a specially-constructed adapter. The diameter of the glass tube was 2 cm and the length was 80 cm. The length of the glass tube was chosen to ensure fully-developed laminar pipe flow at the exit plane of the tube. The determination of the pipe entrance length, L_e , was based on the formula suggested by White (1984)

$$L_e = 0.08 D R_D + 0.7 D \quad (1)$$

where D is the cylinder diameter, and R_D is the flow Reynolds number based on D . The hot-wire sensor was located at the centre of the pipe in the exit plane of the tube. The velocity of the air was calculated from the known flow rate, Q , through the glass tube.

The hot-wire (TSI 1210-T1.5) output voltage was acquired with a constant-temperature anemometer (TSI 1050). The hot-wire was operated at an overheat ratio of 1.8. Data were acquired and processed by an HP200 series computer in conjunction with an HP6944A multi-programmer. The mean velocity over the wire, U_w , was integrated and averaged from the Hagen-Poiseuille profile,

$$U_w = \frac{1}{H} \int_{-H/2}^{+H/2} \frac{2Q(r_0^2 - r^2)}{\pi r_0^4} dr \\ = (2Q/\pi r_0^2)(1 - H^2/12r_0^2) \quad (2)$$

where r_0 is the radius of the glass tube and H is the working length of the hot-wire sensor. The velocity given in equation (2) turns out to be within 1% of the maximum velocity in the pipe for hot-wire sensor and pipe dimensions used in the present experiments. Moreover, the non-uniform temperature distribution along the hot-wire sensor makes the measured velocity even closer to the maximum velocity in the pipe.

The hot-wire calibration was carried out at a fixed temperature, and the calibration data were fitted to a modified King's law,

$$E^2 = A + BU_w^n \quad (3)$$

where E is the anemometer output voltage, and A , B

and n are the best-fit constants. The modified King's law correlated well with the calibration data over the range of 15 to 90 cm/s, even though King's law normally applies for velocities above 25 cm/s where forced convection cooling of the wire dominates.

3. The shedding-frequency method

In this method, the shedding frequency is measured and the velocity is obtained from an empirical SR relationship. Unfortunately, published SR curves in the relevant Reynolds number range for low-speed calibration have discrepancies as high as 20%. The discussion below describes how a continuous SR curve can be used to yield accurate low-speed calibration results.

Flow visualization and hot-wire anemometry were used to develop an accurate calibration method based on cylinder vortex shedding. The vortex-shedding modes of a circular cylinder were examined by smoke-wire flow visualization, and the corresponding SR curves were obtained by hot-wire anemometry. Spanwise flow visualization of the laminar cylinder wake was accomplished with a smoke-wire placed seven diameters downstream and about one diameter off the centre plane of the cylinder. The shedding frequency of the wake was measured with a hot-wire placed ten diameters downstream and two diameters off the centre plane of the cylinder. Vortex-shedding frequencies were obtained from a spectrum analyser (HP3582A).

Both the hot-wire measurements and the smoke-wire flow visualization were performed in a 30 cm × 30 cm × 183 cm suction-type low-speed steady wind tunnel with 25/1 contraction ratio. Two damping tanks (120 cm × 240 cm × 120 cm and 120 cm × 240 cm × 90 cm) lined with acoustic tiles were installed between the test section and the fan to prevent fan noise from reaching the test section. The turbulence level was 0.35% with flow uniformity better than 0.5%. Cylinders with diameters of 1.56 mm and 3.2 mm were made from polished drill rod. The cylinders were stretched across the wind tunnel by weights. Aeroelastic coupling was prevented by choosing the weight to be such that the natural frequency of the cylinder was well above vortex shedding frequencies. The L/D ratios of these cylinders were 95 and 195.

Figure 2(a) shows typical non-parallel vortex shedding for a circular cylinder with no end modifications at $R = 130$. The axis of the vortex filament is not parallel to the axis of the cylinder. The corresponding SR curve is shown in figure 3(a) by open circles. The results indicate a discontinuity in the SR curve at a Reynolds number, R_c , of about 103.

Discontinuities in the SR curves of unmodified cylinders have been observed by many other investigators. Figure 3(a) compares the present SR curve results for non-parallel shedding with those of other investigations. The Reynolds number for the occurrence of the first discontinuity range from 64 to 130. Results from Williamson (1989) and the present investigation indicate that

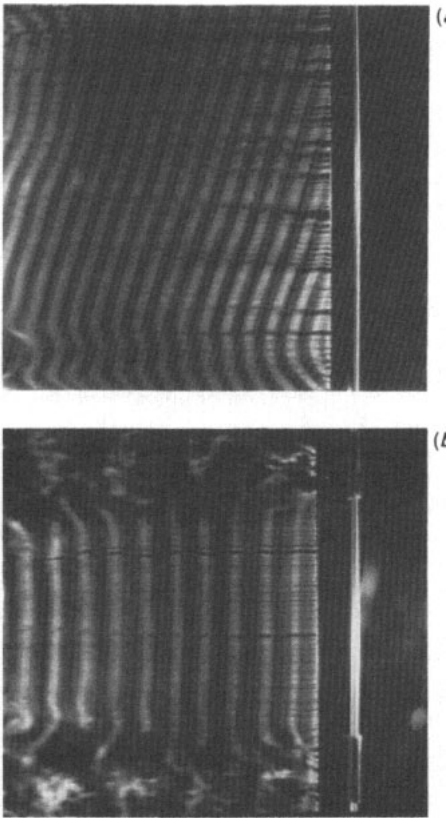


Figure 2. Spanwise flow visualization of the vortex street wake of a circular cylinder, $R=130$ for both: (a) oblique vortex shedding mode for a cylinder with no end modifications; (b) parallel shedding mode due to end cylinders.

the first discontinuity in the SR curve observed by Tritton (1959), Berger and Wille (1972) and Nishioka and Sato (1974) was a result of a discontinuous drop in the shedding angle. Roshko (1954) was the only investigator whose data did not show the first discontinuity for an unmodified cylinder, but the reasons for this are not known. Figure 3(a) also shows that SR curves for non-parallel shedding vary by up to 20%. The collective results clearly indicate that the characteristics of the non-parallel mode cylinder wake are facility dependent. For this reason the shedding-frequency method with non-parallel vortex shedding cannot be used as an accurate low-speed calibration method.

Parallel vortex shedding over a central span of the cylinder has been induced with different end modifications. Williamson (1989) used end discs inclined by 12° – 15° to induce parallel vortex shedding. Hammache and Gharib (1989) generated parallel vortex shedding by positioning two control cylinders upstream and at right angles to the main cylinder.

Figure 2(b) shows flow visualization parallel vortex shedding generated by using cylindrical sleeves at each end of the cylinder (end cylinders). The end cylinders used to promote parallel shedding in the present investigation were similar to those of Eisenlohr and Eckelmann (1989), but without end discs. The present end cylinders had a diameter of $2D$ and a length of $11D$, with a $13D$

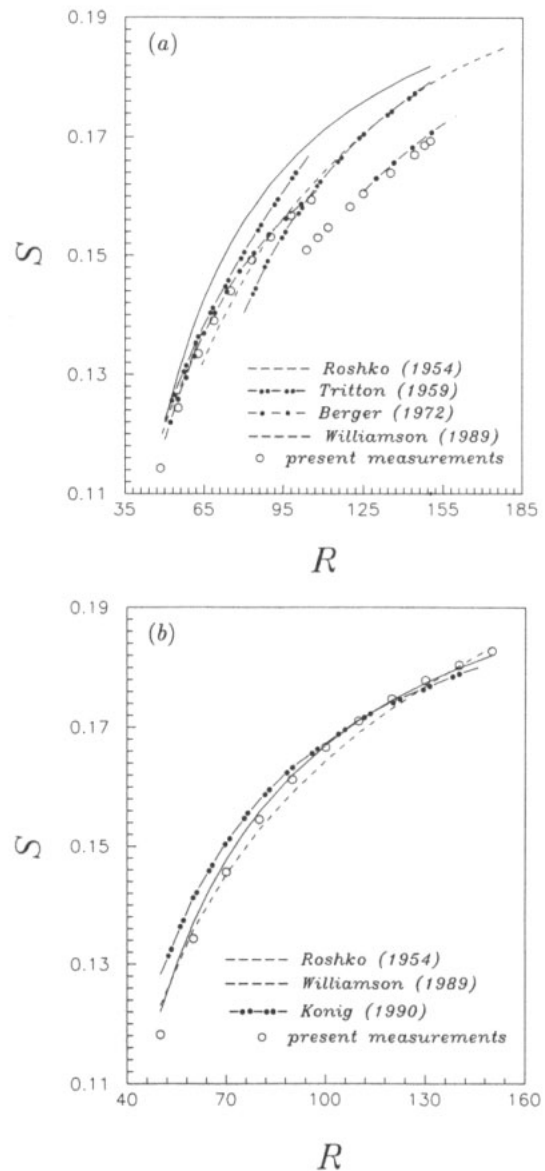


Figure 3. Comparison of the present results with the SR curves of other investigators: (a) discontinuous SR results (except, Roshko (1954)); (b) continuous SR results.

gap between the outer edge of the end cylinders and the walls of the wind tunnel.

Figure 3(b) shows the SR measurements of the parallel vortex-shedding mode. Data from the present investigation are represented by open circles. The data are well fitted by one continuous curve and are within 2% of Roshko's results. The velocity to generate the SR data was determined from a hot-wire placed outside the cylinder wake. The best-fit equation to the present SR data is

$$S = 2.531 \times 10^{-5} R + 0.2102 - 4.332/R. \quad (4)$$

Figure 3(b) also shows that the present results agree well with the SR curves from Williamson (1989), and Konig *et al* (1990). The best-fit line to the continuous

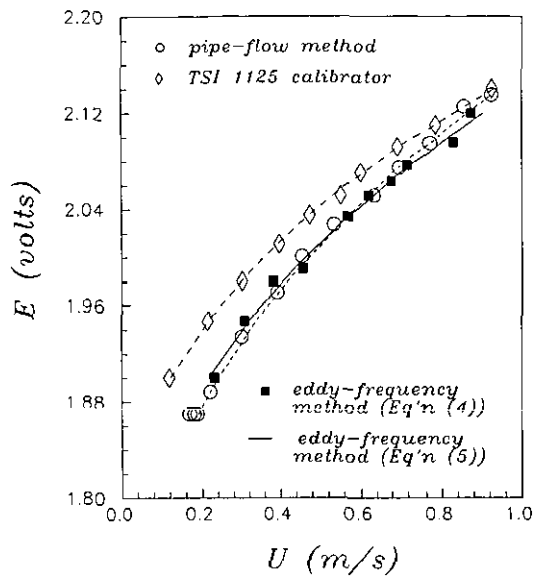


Figure 4. Calibration results of the two improved methods and of the TSI 1125 calibrator in the low velocity range.

results from Williamson, König *et al*, and the present measurements is,

$$S = 1.247 \times 10^{-4} R + 0.1906 - 3.671/R. \quad (5)$$

In summary, the recommended steps to accomplish calibration by the shedding frequency methods are as follows: (i) generate parallel shedding, (ii) measure the shedding frequency, and (iii) use equation (5) to determine the free-stream velocity. The Reynolds number range for the shedding-frequency calibration method is between 50 and 150.

4. Comparisons of the methods and conclusions

Figure 4 shows the comparison of calibration results from three methods: (i) the pipe-flow method, (ii) the shedding-frequency method, and (iii) the TSI 1125 calibrator. Figure 4 reveals that the free-stream velocities determined by the laminar pipe-flow method (open circles) are within $\pm 3\%$ of those obtained by the shedding-frequency method (full line). The uncertainty of pipe-flow calibration results was ± 1.5 cm/s. Figure 4 also shows that the free-stream velocity obtained by

the TSI 1125 calibrator (open rhombuses) was about 4–50% lower than the calibration results of the present methods.

We have described two improved methods for the low speed calibration of hot-wire anemometers. These methods should provide simple, but accurate means for low speed hot-wire calibration.

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